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2201212

Lab8 - Hadoop

Map Reduce

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~/Documents
gruu ~/5 21:52 hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-k.jar -files mapper.py, reducer.py -mapper mapper.py -reducer reducer.py -input /user/wordcount/input/big.txt
-output /user/wordcount/output1
packageJobJar: [/tmp/hadoop-unjar8631378402563199325/] [] /tmp/streamjob1555042320636495985.jar tmpDir=null
2024-11-12 21:52:34,603 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-11-12 21:52:34,729 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-11-12 21:52:35,052 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/gruu/.staging/job_1731427264634_0006
2024-11-12 21:52:35,377 INFO mapred.FileInputFormat: Total input files to process : 1
2024-11-12 21:52:35,426 INFO mapreduce.JobSubmitter: number of splits:2
2024-11-12 21:52:35,575 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1731427264634_0006
2024-11-12 21:52:35,575 INFO mapreduce.JobSubmitter: Executing with tokens: []
2024-11-12 21:52:35,678 INFO conf.Configuration: resource-types.xml not found
2024-11-12 21:52:35,678 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2024-11-12 21:52:35,714 INFO impl.VarClientImpl: Submitted application application_1731427264634_0006
2024-11-12 21:52:35,731 INFO mapreduce.Job: The url to track the job: http://groupc:8088/proxy/application_1731427264634_0006/
2024-11-12 21:52:35,732 INFO mapreduce.Job: Running job: job_1731427264634_0006
2024-11-12 21:52:44,853 INFO mapreduce.Job: Job job_1731427264634_0006 running in uber mode : false
2024-11-12 21:52:44,855 INFO mapreduce.Job: map 0% reduce 0%
2024-11-12 21:52:49,906 INFO mapreduce.Job: map 100% reduce 0%
2024-11-12 21:52:54,937 INFO mapreduce.Job: map 100% reduce 100%
2024-11-12 21:52:55,955 INFO mapreduce.Job: Job job_1731427264634_0006 completed successfully
2024-11-12 21:52:56,087 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=11897862
  FILE: Number of bytes written=24733208
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=6492967
  HDFS: Number of bytes written=736452
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=2
  Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=5408
  Total time spent by all reduces in occupied slots (ms)=2633
  Total time spent by all map tasks (ms)=5408
  Total time spent by all reduce tasks (ms)=2633
  Total vcore-millisecons taken by all map tasks=5408
  Total vcore-millisecons taken by all reduce tasks=2633
  Total megabyte-millisecons taken by all map tasks=5537792
  Total megabyte-millisecons taken by all reduce tasks=2696192
Map-Reduce Framework
  Map input records=128458
  Map output records=1095695
  Map output bytes=9706466
  Map output materialized bytes=11897868
  Input split bytes=204
  Combine input records=0
  Combine output records=0
  Reduce input groups=75161
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=2
  Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=5408
  Total time spent by all reduces in occupied slots (ms)=2633
  Total time spent by all map tasks (ms)=5408
  Total time spent by all reduce tasks (ms)=2633
  Total vcore-millisecons taken by all map tasks=5408
  Total vcore-millisecons taken by all reduce tasks=2633
  Total megabyte-millisecons taken by all map tasks=5537792
  Total megabyte-millisecons taken by all reduce tasks=2696192
Map-Reduce Framework
  Map input records=128458
  Map output records=1095695
  Map output bytes=9706466
  Map output materialized bytes=11897868
  Input split bytes=204
  Combine input records=0
  Combine output records=0
  Reduce input groups=75161
  Reduce shuffle bytes=11897868
  Reduce input records=1095695
  Reduce output records=58553
  Spilled Records=2191390
  Shuffled Maps=2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=31
  CPU time spent (ms)=4820
  Physical memory (bytes) snapshot=1052622848
  Virtual memory (bytes) snapshot=8269520896
  Total committed heap usage (bytes)=754974720
  Peak Map Physical memory (bytes)=360906400
  Peak Map Virtual memory (bytes)=2756481024
  Peak Reduce Physical memory (bytes)=333574144
  Peak Reduce Virtual memory (bytes)=2759372800
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=6492763
File Output Format Counters
  Bytes Written=736452
2024-11-12 21:52:56,087 INFO streaming.StreamJob: Output directory: /user/wordcount/output1
gruu ~/5 21:52
```

Combiner

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~/Documents
gruu ~/Documents 21:50 hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar -files mapper.py,combiner.py -mapper mapper.py -reducer reducer.py -combiner combiner.py -i
nput /user/wordcount/input/big.txt -output /user/wordcount/output3
packageJobJar: [/tmp/hadoop-unjar8963396216941017969/] [] /tmp/streamjob16468463581362007516.jar tmpDir=null
2024-11-12 21:50:45,706 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-11-12 21:50:45,834 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-11-12 21:50:46,163 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/gruu/.staging/job_1731427264634_0005
2024-11-12 21:50:46,502 INFO mapred.FileInputFormat: Total input files to process : 1
2024-11-12 21:50:46,549 INFO mapreduce.JobSubmitter: number of splits:2
2024-11-12 21:50:46,708 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1731427264634_0005
2024-11-12 21:50:46,708 INFO mapreduce.JobSubmitter: Executing with tokens: []
2024-11-12 21:50:46,857 INFO conf.Configuration: resource-types.xml not found
2024-11-12 21:50:46,857 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2024-11-12 21:50:46,902 INFO impl.YarnClientImpl: Submitted application application_1731427264634_0005
2024-11-12 21:50:46,989 INFO mapreduce.Job: The url to track the job: http://groupc:8088/proxy/application_1731427264634_0005/
2024-11-12 21:50:46,910 INFO mapreduce.Job: Running job: job_1731427264634_0005
2024-11-12 21:50:51,986 INFO mapreduce.Job: Job job_1731427264634_0005 running in uber mode : false
2024-11-12 21:50:51,988 INFO mapreduce.Job: map 0% reduce 0%
2024-11-12 21:50:56,041 INFO mapreduce.Job: map 100% reduce 0%
2024-11-12 21:51:01,072 INFO mapreduce.Job: map 100% reduce 100%
2024-11-12 21:51:01,082 INFO mapreduce.Job: Job job_1731427264634_0005 completed successfully
2024-11-12 21:51:01,180 INFO mapreduce.Job: Counters: 94
  File System Counters
    FILE: Number of bytes read=1045153
    FILE: Number of bytes written=3029707
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=6492967
    HDFS: Number of bytes written=736452
    HDFS: Number of read operations=11
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=2
    Launched reduce tasks=1
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=4975
    Total time spent by all reduces in occupied slots (ms)=2010
    Total time spent by all map tasks (ms)=4975
    Total time spent by all reduce tasks (ms)=2010
    Total vcore-milliseonds taken by all map tasks=4975
    Total vcore-milliseonds taken by all reduce tasks=2010
    Total megabyte-milliseonds taken by all map tasks=5094400
    Total megabyte-milliseonds taken by all reduce tasks=2058240
  Map-Reduce Framework
    Map input records=128458
    Map output records=1095695
    Map output bytes=9706466
    Map output materialized bytes=1045159
    Input split bytes=204
    Combine input records=1095695
    Combine output records=72982
    Reduce input groups=70567
```

```
~/Documents
    HDFS: Number of read operations=11
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=2
    Launched reduce tasks=1
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=4975
    Total time spent by all reduces in occupied slots (ms)=2010
    Total time spent by all map tasks (ms)=4975
    Total time spent by all reduce tasks (ms)=2010
    Total vcore-milliseonds taken by all map tasks=4975
    Total vcore-milliseonds taken by all reduce tasks=2010
    Total megabyte-milliseonds taken by all map tasks=5094400
    Total megabyte-milliseonds taken by all reduce tasks=2058240
  Map-Reduce Framework
    Map input records=128458
    Map output records=1095695
    Map output bytes=9706466
    Map output materialized bytes=1045159
    Input split bytes=204
    Combine input records=1095695
    Combine output records=72982
    Reduce input groups=70567
    Reduce shuffle bytes=1045159
    Reduce input records=72982
    Reduce output records=58553
    Spilled Records=145964
    Shuffled Maps =2
    Failed Shuffles=0
    Merged Map outputs=2
    GC time elapsed (ms)=27
    CPU time spent (ms)=4540
    Physical memory (bytes) snapshot=980471808
    Virtual memory (bytes) snapshot=8267116544
    Total committed heap usage (bytes)=754974720
    Peak Map Physical memory (bytes)=360838112
    Peak Map Virtual memory (bytes)=2760011776
    Peak Reduce Physical memory (bytes)=262242304
    Peak Reduce Virtual memory (bytes)=2757267456
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=6492763
  File Output Format Counters
    Bytes Written=736452
2024-11-12 21:51:01,180 INFO streaming.StreamJob: Output directory: /user/wordcount/output3
gruu ~/Documents 21:51
```

K-means

```
~/Documents
gruu ~/ 21:47 hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar -files mapper_K.py,reducer_K.py -mapper mapper_K.py -reducer reducer_K.py -input /user/wordcount/input/iris.txt -output /user/wordcount/output2
packageJobJar: [/tmp/hadoop-unjar1664550333767862639/] [] /tmp/streamjob5766156634194060631.jar tmpDir=null
2024-11-12 21:48:30,508 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-11-12 21:48:30,639 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2024-11-12 21:48:31,003 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/gruu/.staging/job_1731427264634_0004
2024-11-12 21:48:31,317 INFO mapred.FileInputFormat: Total input files to process : 1
2024-11-12 21:48:31,367 INFO mapreduce.JobSubmitter: number of splits:2
2024-11-12 21:48:31,521 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1731427264634_0004
2024-11-12 21:48:31,521 INFO mapreduce.JobSubmitter: Executing with tokens: []
2024-11-12 21:48:31,626 INFO conf.Configuration: resource-types.xml not found
2024-11-12 21:48:31,626 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2024-11-12 21:48:31,662 INFO impl.YarnClientImpl: Submitted application application_1731427264634_0004
2024-11-12 21:48:31,681 INFO mapreduce.Job: The url to track the job: http://groupc:8088/proxy/application_1731427264634_0004/
2024-11-12 21:48:31,682 INFO mapreduce.Job: Running job: job_1731427264634_0004
2024-11-12 21:48:35,745 INFO mapreduce.Job: Job job_1731427264634_0004 running in uber mode : false
2024-11-12 21:48:35,746 INFO mapreduce.Job: map 0% reduce 0%
2024-11-12 21:48:39,797 INFO mapreduce.Job: map 100% reduce 0%
2024-11-12 21:48:44,829 INFO mapreduce.Job: map 100% reduce 100%
2024-11-12 21:48:44,839 INFO mapreduce.Job: Job job_1731427264634_0004 completed successfully
2024-11-12 21:48:44,938 INFO mapreduce.Job: Counters: 94
  File System Counters
    FILE: Number of bytes read=1956
    FILE: Number of bytes written=941447
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=7039
    HDFS: Number of bytes written=117
    HDFS: Number of read operations=11
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=2
    Launched reduce tasks=1
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=3658
    Total time spent by all reduces in occupied slots (ms)=2122
    Total time spent by all map tasks (ms)=3658
    Total time spent by all reduce tasks (ms)=2122
    Total vcore-milliseonds taken by all map tasks=3658
    Total vcore-milliseonds taken by all reduce tasks=2122
    Total megabyte-milliseonds taken by all map tasks=3745792
    Total megabyte-milliseonds taken by all reduce tasks=2172928
  Map-Reduce Framework
    Map input records=152
    Map output records=150
    Map output bytes=1650
    Map output materialized bytes=1962
    Input split bytes=206
    Combine input records=0
    Combine output records=0
    Reduce input groups=116
```

```
~/Documents
    Launched map tasks=2
    Launched reduce tasks=1
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=3658
    Total time spent by all reduces in occupied slots (ms)=2122
    Total time spent by all map tasks (ms)=3658
    Total time spent by all reduce tasks (ms)=2122
    Total vcore-milliseonds taken by all map tasks=3658
    Total vcore-milliseonds taken by all reduce tasks=2122
    Total megabyte-milliseonds taken by all map tasks=3745792
    Total megabyte-milliseonds taken by all reduce tasks=2172928
  Map-Reduce Framework
    Map input records=152
    Map output records=150
    Map output bytes=1650
    Map output materialized bytes=1962
    Input split bytes=206
    Combine input records=0
    Combine output records=0
    Reduce input groups=116
    Reduce shuffle bytes=1962
    Reduce input records=150
    Reduce output records=3
    Spilled Records=300
    Shuffled Maps =2
    Failed Shuffles=0
    Merged Map outputs=2
    GC time elapsed (ms)=28
    CPU time spent (ms)=1490
    Physical memory (bytes) snapshot=969248768
    Virtual memory (bytes) snapshot=8263667712
    Total committed heap usage (bytes)=754974720
    Peak Map Physical memory (bytes)=357679104
    Peak Map Virtual memory (bytes)=2754617344
    Peak Reduce Physical memory (bytes)=258424832
    Peak Reduce Virtual memory (bytes)=2754940928
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=6833
  File Output Format Counters
    Bytes Written=117
2024-11-12 21:48:44,939 INFO streaming.StreamJob: Output directory: /user/wordcount/output2

gruu ~/ 21:48 hdfs dfs -cat /user/wordcount/output2/*
0,5.076923076923076,3.5461538461538464
1,5.588709677419357,2.833870967741936
2,6.77551020408163,2.940816326530612

gruu ~/ 21:50
```